

# Technical Log

# Music Analysis Tool

## Overview

Import an audio file and return key signature, time signature, and tempo.

## Design Limitations

| Feature        | Status   | Notes                                                                                            |
|----------------|----------|--------------------------------------------------------------------------------------------------|
| Tempo          | Reliable | Less accurate on songs with large tempo shifts                                                   |
| Key            | Reliable | Standard algorithm struggles with minor keys and modes; the custom system improves this          |
| Time Signature | Limited  | Overall unreliable, especially for synthesized or melodic audio. May try to implement the future |

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## Known Issues

- Krumhansl-Schmuckler frequently confuses relative major/minor keys
  - Time signature detection is fundamentally difficult without strong rhythmic accents
  - Mixolydian and Phrygian are typically identified together, but this is one-directional (Mixolydian scale gets flagged as Phrygian, but Phrygian scales are correctly identified)
  - Locrian has issues being identified using a pure scale.
  - Lack of songs to test all modes; accuracy using real audio is not currently guaranteed.
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## Custom Key Detection

Built a custom profile-based key estimator using Pearson correlation. Returns top 3 results with confidence scores—Outperforms music21's built-in Krumhansl-Schmuckler on several test cases, particularly for harmonic minor keys. Currently implements 108 profiles across 9 scale types: Major, Natural Minor, Harmonic Minor, Dorian, Phrygian, Lydian, Mixolydian, Phrygian Dominant, and Locrian (12 keys each).

**Future extensions:** Chord progression generation, Chord progression identification, Time signature identification



# Dev Log

## Development Log

### 5/29/26 — Initial Setup & Core Features

#### Setup

- Created `test.py` to verify library imports and versions
- Resolved Python interpreter mismatch between terminal and VS Code

#### Features Implemented

- Tempo detection via `librosa.beat.beat_track` + interval method as backup
- Smoothed tempo curve over time using `scipy.ndimage.uniform_filter1d`
- Chromagram visualization — heatmap and bar chart
- Key detection via Krumhansl-Schmuckler (music21)
- Custom key detection system — 36 profiles across major, natural minor, and harmonic minor

### 5/30/26 — Fixing Errors and Extending Features

#### Errors

- Resolved Logic error within the base `c_natural_minor` that gave weight to A instead of G#
- Updated logic for custom key identities: 1st, 3rd, and 5th notes are given 6.0 weighting, other notes in the mode are given 3.0, and the rest are given 1.0

#### Features Implemented

- Added new sample scales to test differences between librosa detection and my custom-built key detection
- Dorian, Phrygian, Locrian, Lydian, Mixolydian, and Phrygian Dominant were implemented into custom key detection
- Adjusted Locrian profile — diminished 5th (F#), weight reduced from 6.0 to 3.0 to reflect the instability of the interval. Correctly identifies C Locrian after adjustment. However, this adjustment then identifies C Dorian as C Locrian. Reverted to 6.0
- Updated Key Signatures to use the standard naming, although some naming for modes may be incorrect.

#### Current Limitations

- Mixolydian/Phrygian Ambiguity – Modes with the same parent scale (e.g., C Mixolydian and A Phrygian) produce nearly identical scores when tested using a scale
- Locrian – Unreliable detection due to diminished 5th
- **General note** — Chromagram-based key detection fundamentally cannot detect tonal center, only pitch content.
- Lack of proper songs to test all modes.

# Test Results

### 5/29/26 - Initial Tests (Major, Minor, Harmonic Only)

| Song                       | Known Key | Known BPM | Custom BPM      | Custom Key                       | K-S Result               | Correct? |
|----------------------------|-----------|-----------|-----------------|----------------------------------|--------------------------|----------|
| 老張 - No Party for Cao Dong | E Major   | 135       | 135.31 - 135.99 | E Major                          | G# Minor<br>Alt: B Major | ✓        |
| Tank! - The Seatbelts      | C Minor   | 137-140   | 135.99 - 137.52 | C Minor<br>Alt: C Harmonic Minor | C Major<br>Alt: C Minor  | ✓        |

#### Notes:

- Custom system outperforms K-S on both tests
- Tank! - The Seatbelts
  - Key is ambiguous between sources (Spotify vs Wikipedia)
  - Lower confidence scores overall reflect the harmonic complexity of the jazz arrangement.
  - Tempos are slightly off from 140.
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- 老張 - No Party For Cao Dong
  - Custom system correctly identified E major with high confidence (0.846).
  - Krumhansl-Schmuckler returned G# minor, the relative minor of E major.

### 5/30/26

| Song                                       | Known Key    | Known BPM | Custom BPM     | Custom Key                             | K-S Result              | Correct? |
|--------------------------------------------|--------------|-----------|----------------|----------------------------------------|-------------------------|----------|
| Love Me Again - John Newman                | G Dorian     | 127       | 126 - 126.42   | G Dorian<br>Alt: G Minor               | G Minor<br>Alt: G Major | ✓        |
| Can You Feel the Love Tonight - Elton John | B ♭ Major    | 126       | 129.2 - 127.36 | B ♭ Major                              | B ♭ Major               | ✓        |
| Don't Tell Me - Madonna                    | D Mixolydian | 100       | 99.38 - 99.87  | D Lydian<br>Alt: D Mixolydian, D Major | D Major                 | ✗        |

**Notes:**

- There does not seem to be any issues identifying Dorian, Major, Minor, Harmonic Minor.
- Some of the keys need to be updated to convention: i.e., B  $\flat$  Major instead of A $\sharp$  Major (Which is what is currently being used)
- Don't Tell Me
  - D Lydian, D Mixolydian, and D Major are returned. This is a bit confusing, as Lydian and Mixolydian do not really have overlap.
  - Could be the Chromagram identifying A as G $\sharp$  incorrectly, but I'm doubtful based on the heatmap.
- Overall, I need more songs with different keys to test accuracy.